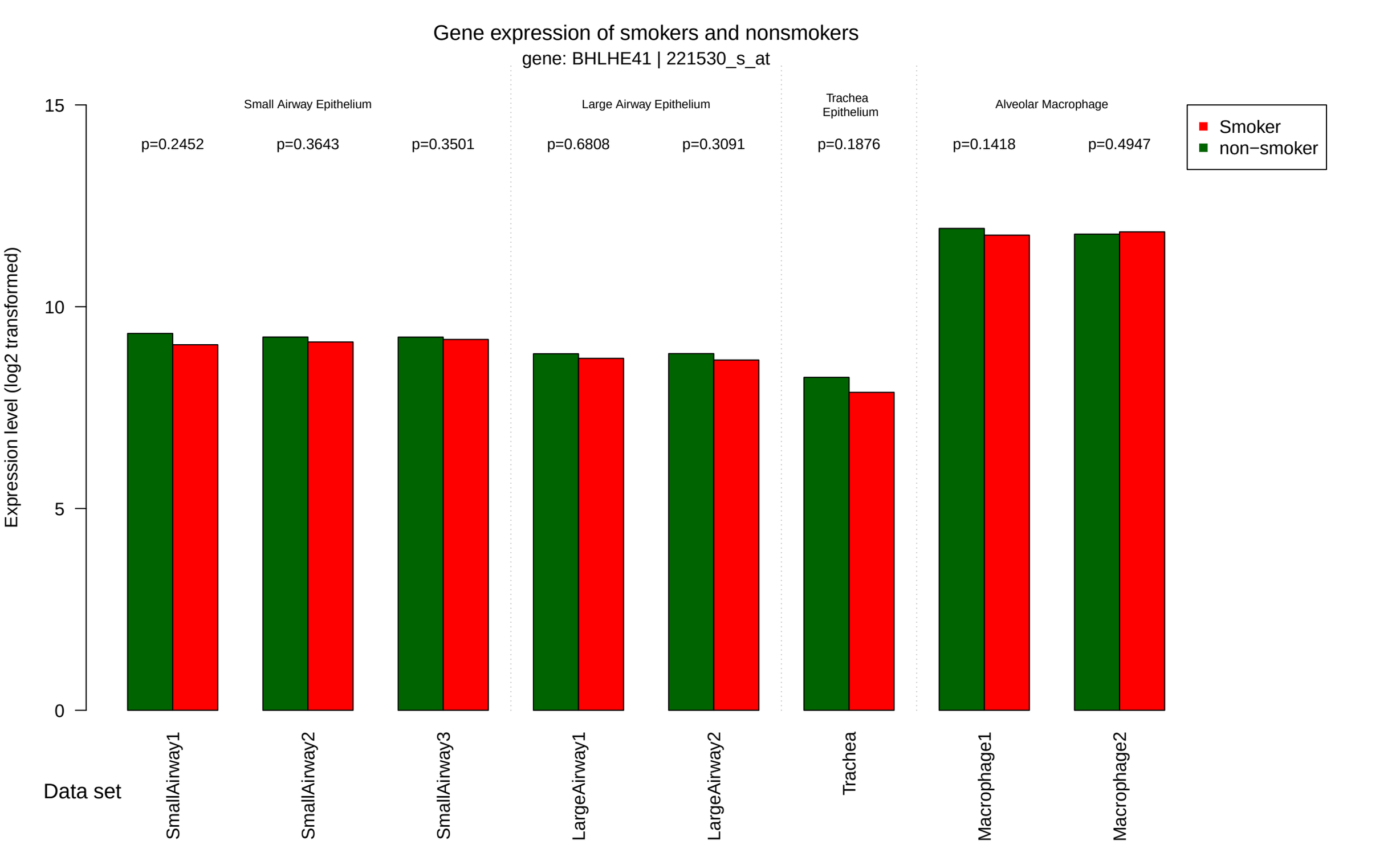
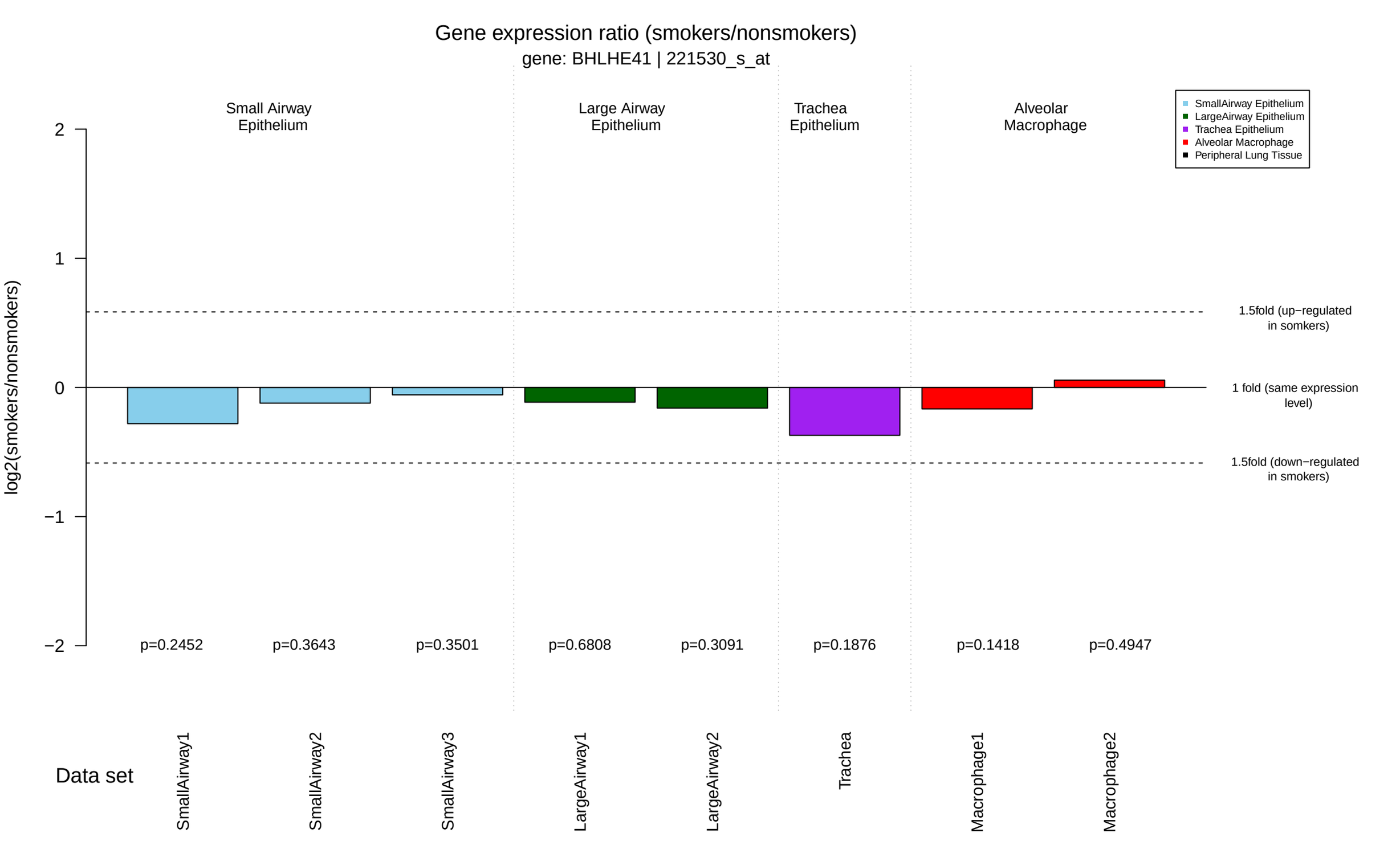
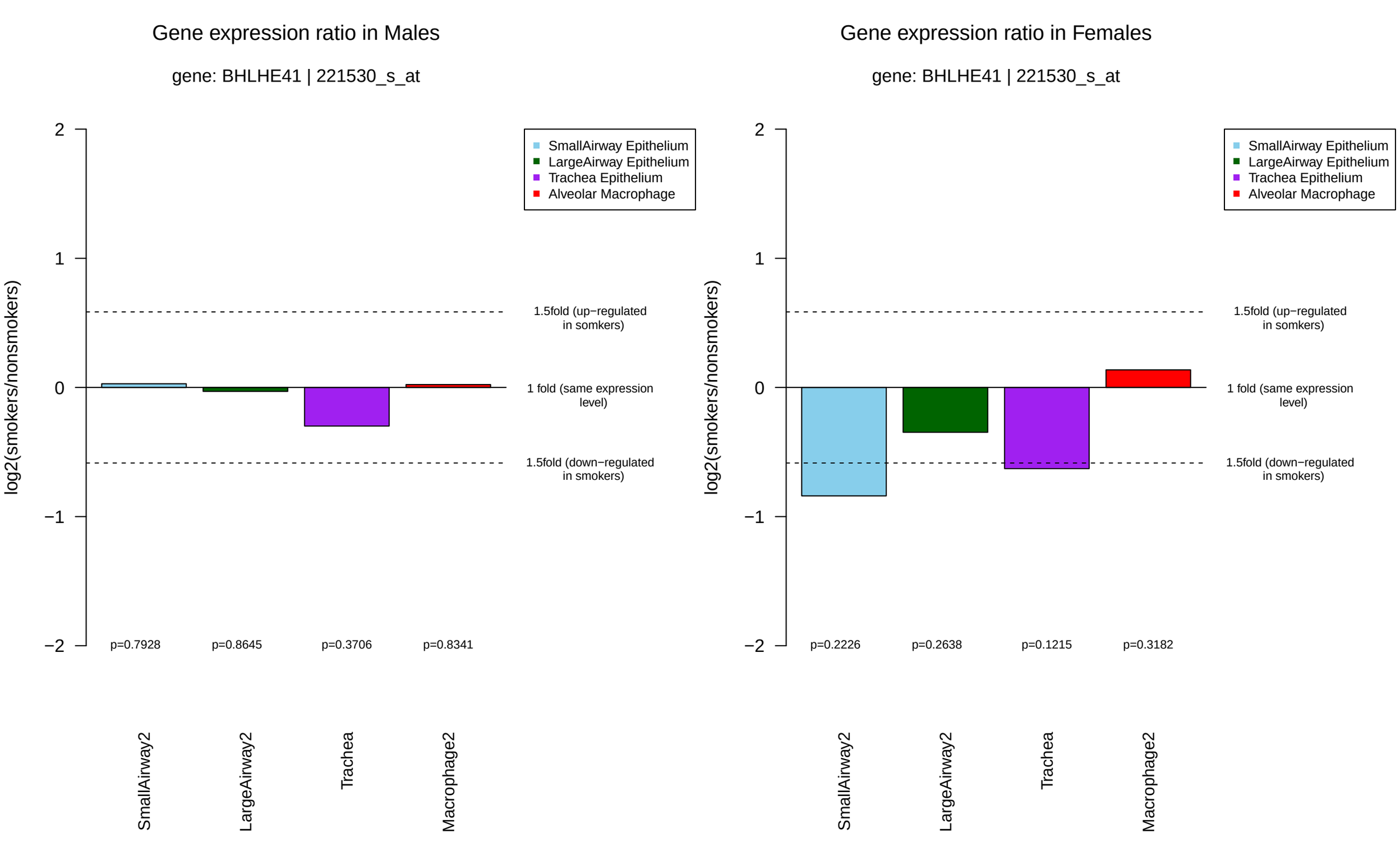
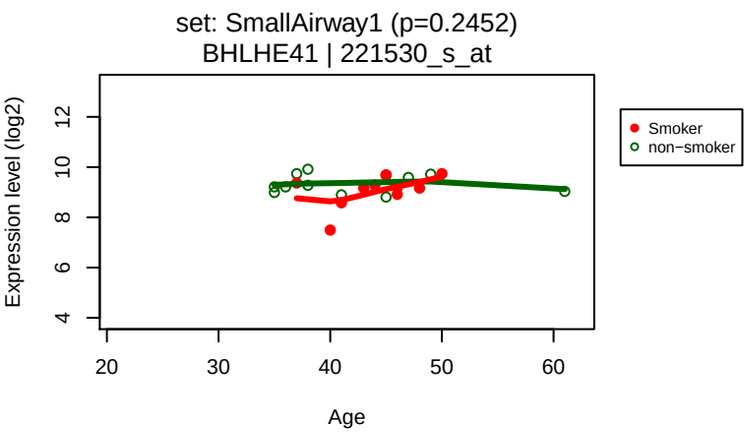
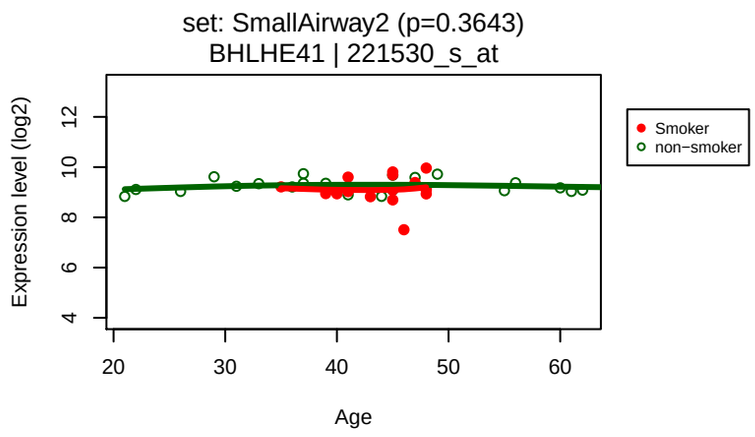
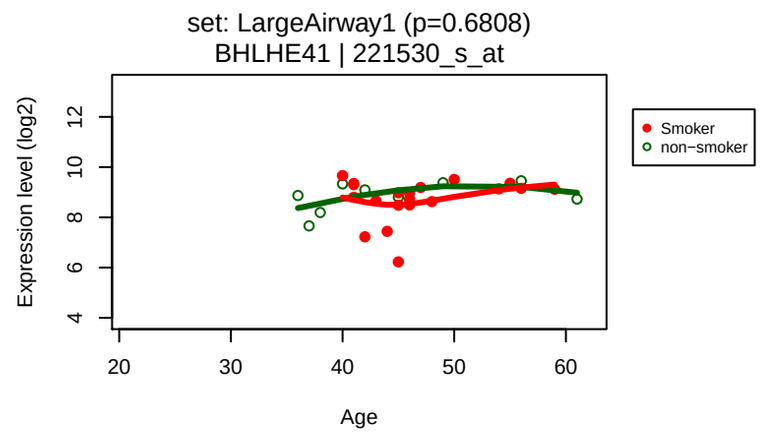

Gene expression of smokers and nonsmokers
gene: BHLHE41 | 221530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2452
p=0.3643
p=0.3501
p=0.6808
p=0.3091
p=0.1876
p=0.1418
p=0.4947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

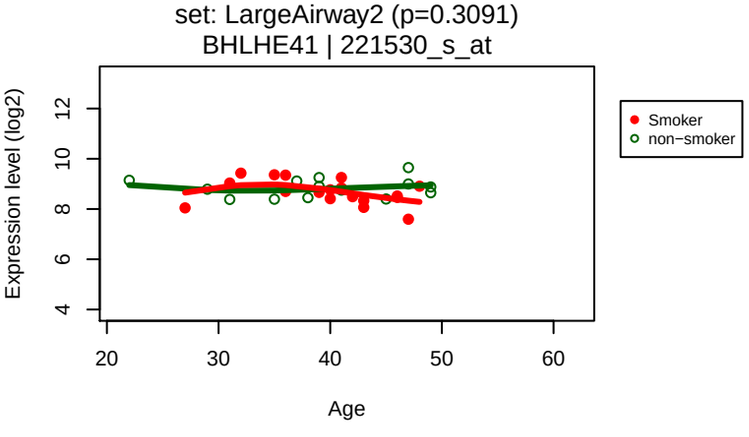

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE41 | 221530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2452
p=0.3643
p=0.3501
p=0.6808
p=0.3091
p=0.1876
p=0.1418
p=0.4947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

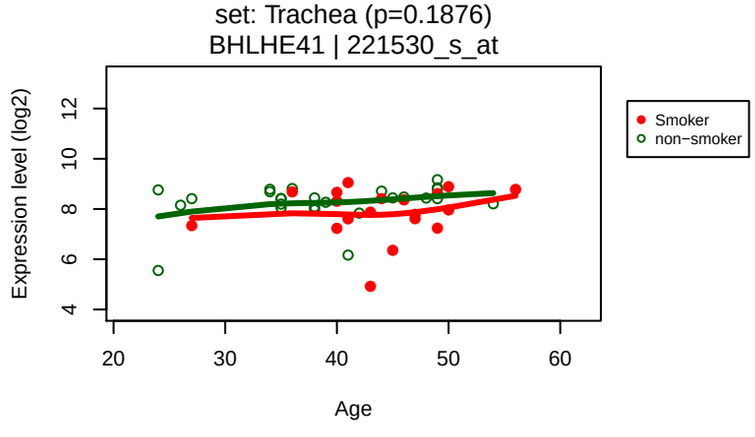

Gene expression ratio in Males
gene: BHLHE41 | 221530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7928
p=0.8645
p=0.3706
p=0.8341
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE41 | 221530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2226
p=0.2638
p=0.1215
p=0.3182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

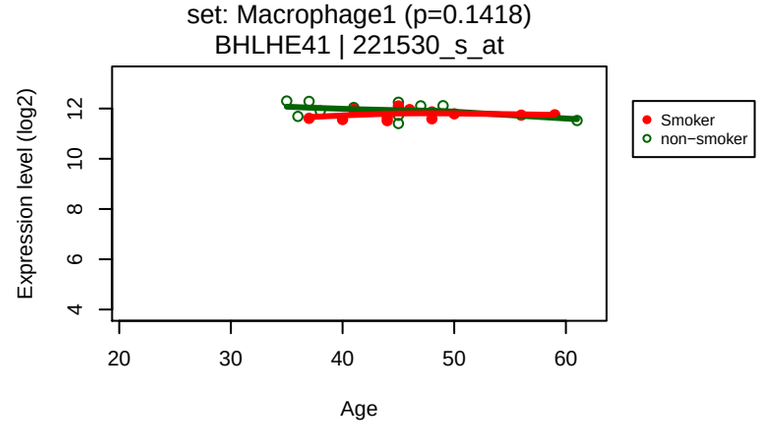

set: SmallAirway1 (p=0.2452)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker

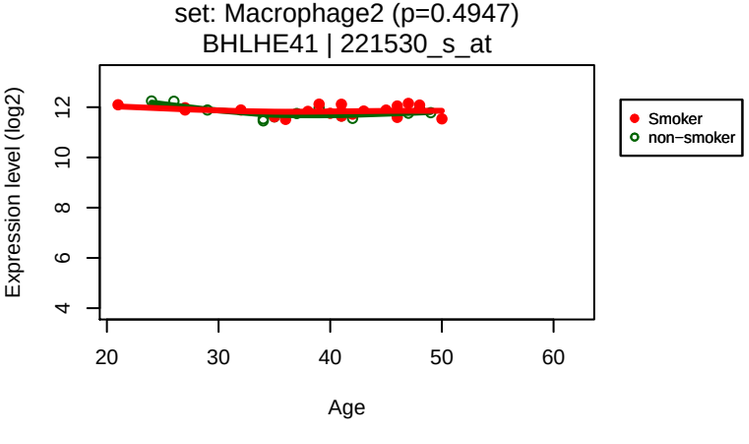

set: SmallAirway2 (p=0.3643)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker

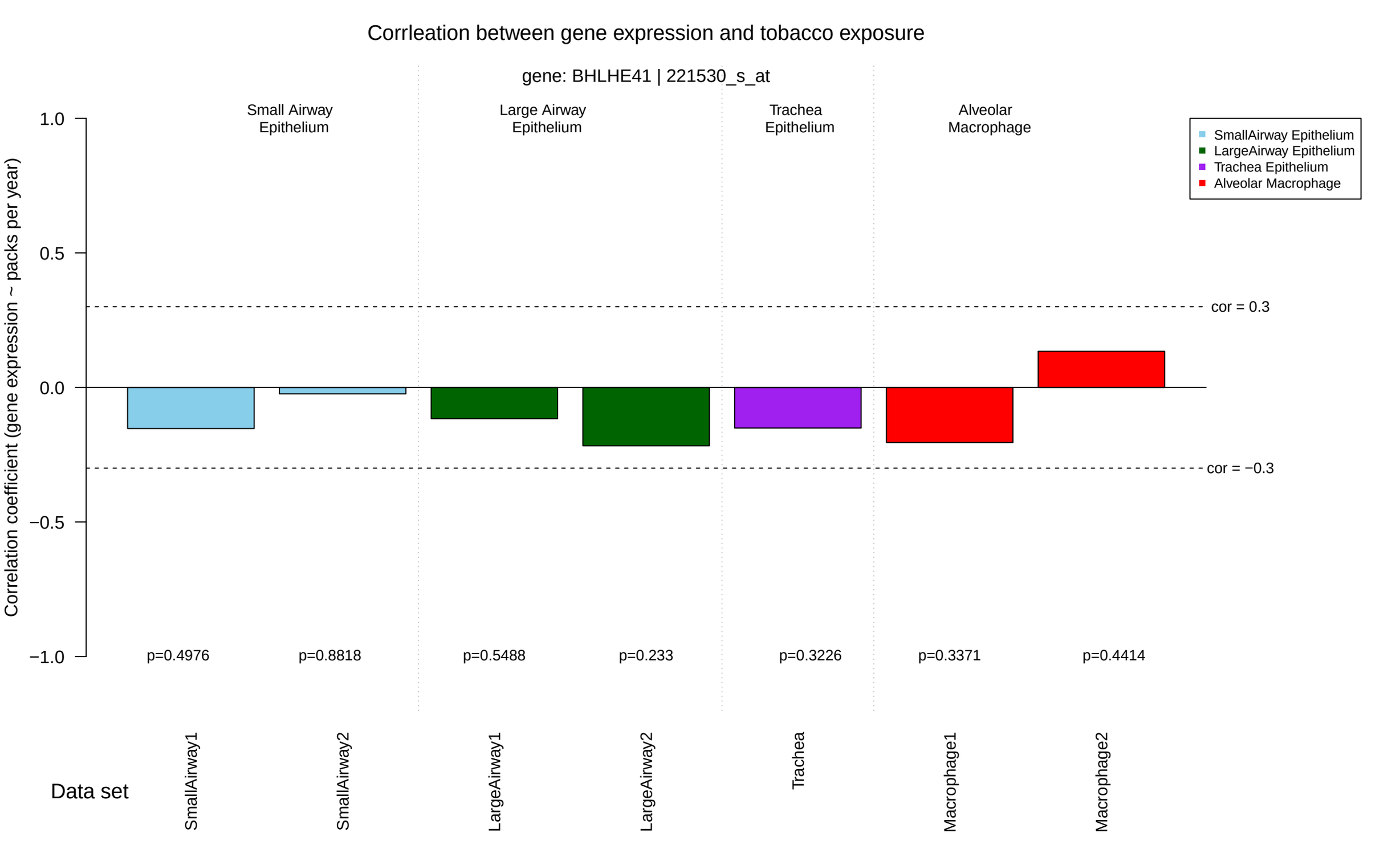

set: LargeAirway1 (p=0.6808)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3091)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1876)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1418)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4947)
BHLHE41 | 221530_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE41 | 221530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4976
p=0.8818
p=0.5488
p=0.233
p=0.3226
p=0.3371
p=0.4414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2